

**MEDICAL  
APPOINTMENT NO  
SHOW  
CLEANING AND  
ANALYSIS**

# **SUMMARY**

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4. CLEANING
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# **INTRODUCTION**

This project analyses the Medical Appointment No Shows dataset from Kaggle (<https://www.kaggle.com/datasets/joniarroba/noshowappointments/data>). The dataset contains information about patients' appointments, as well as their medical history and health conditions, their neighborhood, and whether or not they showed up for their appointment.

The objective is to analyze the behavior of patients who make medical appointments and to understand if certain parameters can influence the reasons for their absence.

The data was cleaned and analyzed using Python and several visualizations were created to support the findings.

# **DATASET OVERVIEW**

This dataset contains 110.527 medical appointments, and 14 associated variables :

- Patient Id
- Appointment ID
- Gender
- Scheduled Day
- Appointment Day
- Age
- Neighbourhood
- Scholarship
- Hypertension
- Diabetes
- Alcoholism
- Handicap
- SMS\_received
- No-show

# **OBJECTIVES**

- What is the overall no-show rate?
- Does age influence attendance?
- Does gender influence attendance?
- Do SMS reminders reduce no-shows?
- Which neighbourhoods have the highest no-show rates?
- Does waiting time between scheduling and appointment affect attendance?
- Do days of the week affect attendance?
- Do months affect attendance?

# CLEANING

The dataset was imported into Google Colab to clean the data using the Python programming language. Since this database was available on Kaggle, it was imported directly from that platform.

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import kagglehub

path =
kagglehub.dataset_download("joniarroba/noshowappointments")

import os
print(os.listdir(path))

import glob
files = glob.glob(path + "/*.csv")
```

```
csv_file = os.path.join(path, 'KaggleV2-May-2016.csv')
df = pd.read_csv(csv_file)
df.head()
```

```
list(df.columns)
```

```
#Rename columns with wrong syntaxes
df = df.rename(columns={'Hipertension': 'Hypertension',
'Handcap': 'Handicap'})
```

```
#Check duplicate values
df.duplicated().sum()
np.int64(0)
```

```
#Missing values
df.isnull().sum()
```

|                |   |
|----------------|---|
|                | 0 |
| PatientId      | 0 |
| AppointmentID  | 0 |
| Gender         | 0 |
| ScheduledDay   | 0 |
| AppointmentDay | 0 |
| Age            | 0 |
| Neighbourhood  | 0 |
| Scholarship    | 0 |
| Hypertension   | 0 |
| Diabetes       | 0 |
| Alcoholism     | 0 |
| Handicap       | 0 |
| SMS_received   | 0 |
| No-show        | 0 |

```
df.info()
```

```
#Convert dates
df['Scheduled Day'] = pd.to_datetime(df['ScheduledDay'])

df["Appointment Day"] = pd.to_datetime(df['AppointmentDay'])

df['Time'] = pd.to_datetime(df['ScheduledDay']).dt.time
```

```
#Extract days of the week, months and week-ends
#Days of the week
df['Appointment day'] = df['Appointment Day'].dt.day_name()
#Month
df['Appointment month'] = df['Appointment
Day'].dt.month_name()
```

```
#Week-ends
df['Appointment on Weekend'] = df['Appointment
Day'].dt.weekday >= 5
```

```
#Time elapsed between scheduled day and appointment day
df["Waiting Days"] = (df['Appointment Day'].dt.normalize() -
df['Scheduled Day'].dt.normalize())
df['Waiting Days'] = df['Waiting Days'].dt.days
```

```
#Check negative values
df[df['Waiting Days'] < 0]
```

```
#Drop negative values
df = df[df['Waiting Days'] >= 0]
```

```
df['Waiting Days'].describe()
```

|              | Waiting Days  |
|--------------|---------------|
| <b>count</b> | 110522.000000 |
| <b>mean</b>  | 10.184253     |
| <b>std</b>   | 15.255115     |
| <b>min</b>   | 0.000000      |
| <b>25%</b>   | 0.000000      |
| <b>50%</b>   | 4.000000      |
| <b>75%</b>   | 15.000000     |
| <b>max</b>   | 179.000000    |

```
#Check extreme waiting days
df[df['Waiting Days'] > 100][['PatientId', 'Scheduled Day',
'Appointment Day', 'Waiting Days']]
```



|            | PatientId    | Scheduled Day                | Appointment Day              | Waiting Days |
|------------|--------------|------------------------------|------------------------------|--------------|
| 953        | 4.616858e+12 | 2016-01-05<br>15:02:46+00:00 | 2016-04-29<br>00:00:00+00:00 | 115          |
| 954        | 1.423329e+12 | 2016-01-05<br>11:53:44+00:00 | 2016-04-29<br>00:00:00+00:00 | 115          |
| 955        | 1.596183e+14 | 2016-01-11<br>10:20:41+00:00 | 2016-04-29<br>00:00:00+00:00 | 109          |
| 956        | 3.136481e+14 | 2016-01-11<br>09:57:10+00:00 | 2016-04-29<br>00:00:00+00:00 | 109          |
| 957        | 9.189694e+13 | 2016-01-11<br>09:08:10+00:00 | 2016-04-29<br>00:00:00+00:00 | 109          |
| ...        | ...          | ...                          | ...                          | ...          |
| 102<br>872 | 2.552524e+14 | 2015-12-15<br>10:15:50+00:00 | 2016-06-08<br>00:00:00+00:00 | 176          |
| 110<br>296 | 7.682333e+13 | 2016-02-05<br>10:37:16+00:00 | 2016-06-06<br>00:00:00+00:00 | 122          |
| 110<br>321 | 1.388630e+11 | 2016-02-11<br>16:14:32+00:00 | 2016-06-01<br>00:00:00+00:00 | 111          |
| 110<br>335 | 3.747953e+11 | 2016-02-11<br>16:39:26+00:00 | 2016-06-01<br>00:00:00+00:00 | 111          |
| 110<br>351 | 4.483937e+13 | 2016-02-19<br>14:12:52+00:00 | 2016-06-01<br>00:00:00+00:00 | 103          |

138 rows × 4 columns

```
#Check gender values
df['Gender'].value_counts()
```

|        | count |
|--------|-------|
| Gender |       |
| F      | 71837 |
| M      | 38685 |

```
#Check no show values
df['No-show'].value_counts()
```

|         | count |
|---------|-------|
| No-show |       |
| No      | 88208 |
| Yes     | 22314 |

```
#Check ages
df['Age'].value_counts()
```

|     | count |
|-----|-------|
| Age |       |
| 0   | 3539  |
| 1   | 2273  |
| 52  | 1746  |
| 49  | 1652  |
| 53  | 1651  |
| ... | ...   |
| 115 | 5     |
| 100 | 4     |
| 102 | 2     |
| 99  | 1     |
| -1  | 1     |

```
#Drop negative ages
df = df[df['Age']>=0]
```



# EXPLANATORY ANALYSIS

*“What is the overall no-show rate?”*

```
#Overall no-show rate
df['No-show'].value_counts(normalize=True)['Yes'] * 100

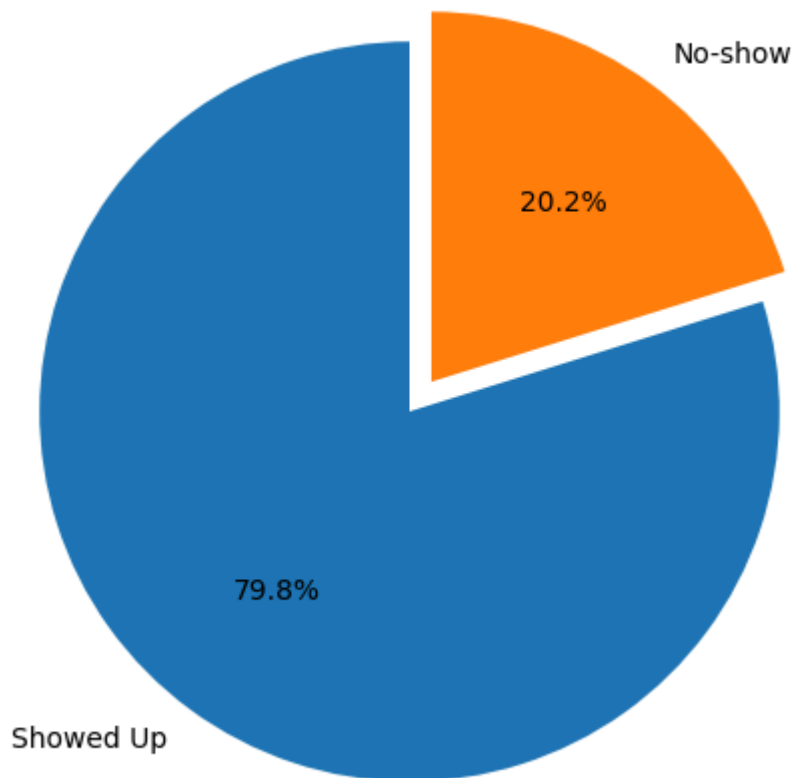
np.float64(20.18982817745044)
```

```
#Pie chart
no_show_counts = df['No-show'].value_counts()

plt.figure(figsize=(6,6))
plt.pie(
    no_show_counts,
    labels=['Showed Up', 'No-show'],
    autopct='%1.1f%%',
    startangle=90,
    explode=(0, 0.1)
)

plt.title('Overall Appointment Attendance', fontweight='bold')
plt.show()
```

## Overall Appointment Attendance



The analysis shows that only a small proportion of patients do not show up for their appointments, which represents one scheduled appointment out of five.

*“Does age influence attendance?”*

```
df['Age'].describe()
```

| Age   |               |
|-------|---------------|
| count | 110521.000000 |
| mean  | 37.089386     |
| std   | 23.109885     |
| min   | 0.000000      |
| 25%   | 18.000000     |

|            |            |
|------------|------------|
| <b>50%</b> | 37.000000  |
| <b>75%</b> | 55.000000  |
| <b>max</b> | 115.000000 |

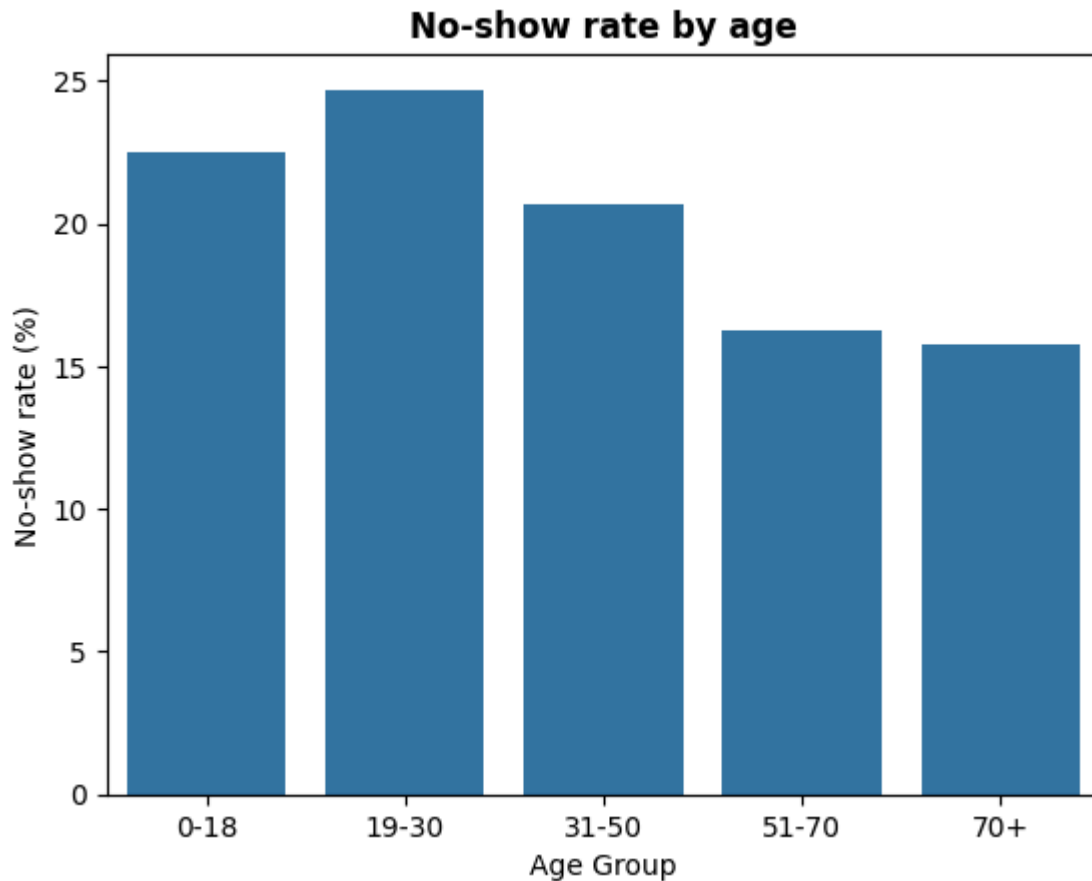
```
#Create age groups
df['Age_group'] = pd.cut(df['Age'], bins=[0, 18, 30, 50, 70, 120], labels=['0-18', '19-30', '31-50', '51-70', '70+'])
```

```
#Age influence
df['No_show_values'] = df['No-show'].map({'No':0, 'Yes': 1})

age_influence =
df.groupby('Age_group')['No_show_values'].mean()*100
age_influence
```

| Age_group    | No_show_values |
|--------------|----------------|
| <b>0-18</b>  | 22.534155      |
| <b>19-30</b> | 24.703215      |
| <b>31-50</b> | 20.701707      |
| <b>51-70</b> | 16.271881      |
| <b>70+</b>   | 15.804832      |

```
#Bar chart
sns.barplot(x=age_influence.index, y=age_influence.values)
plt.xlabel('Age Group')
plt.ylabel('No-show rate (%)')
plt.title('No-show rate by age', fontweight='bold')
plt.show()
```



The results indicate that no-show rates are higher among younger patients and decrease progressively with age. Patients under 30 years old are the most likely to miss their appointments, while attendance rates improve steadily among older age groups.

Although the dataset does not provide information about the reasons behind this behavior, one possible explanation is that older individuals may be more aware of the importance of medical follow-up due to age-related health concerns. Younger patients, who are often in better overall health, may be less likely to prioritize medical appointments.

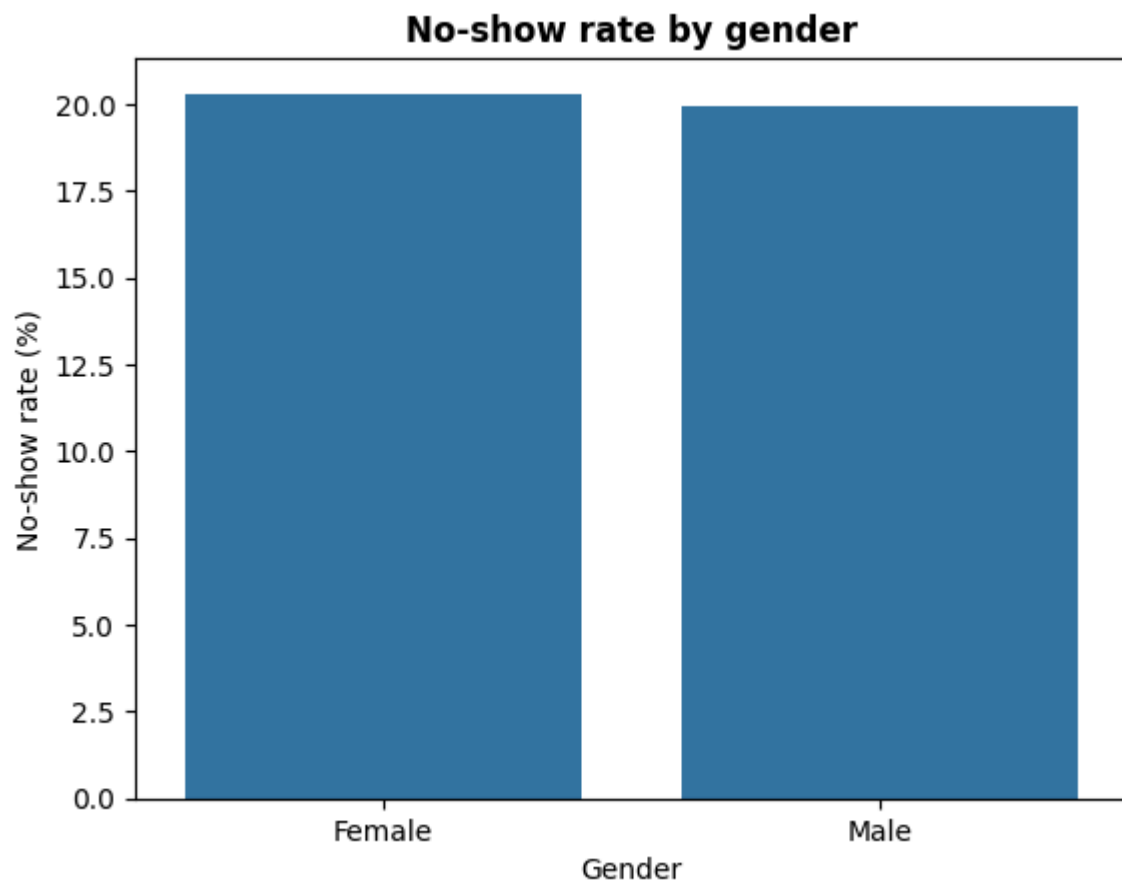
*“Does gender influence attendance?”*

```
df['Gender_label'] = df['Gender'].map({'M': 'Male', 'F': 'Female'})
gender_influence =
df.groupby('Gender_label')['No_show_values'].mean()*100
gender_influence
```

No\_show\_value  
s

| Gender_label |           |
|--------------|-----------|
| Female       | 20.311543 |
| Male         | 19.963810 |

```
#Bar chart
sns.barplot(x=gender_influence.index,
y=gender_influence.values)
plt.xlabel('Gender')
plt.ylabel('No-show rate (%)')
plt.title('No-show rate by gender', fontweight='bold')
plt.show()
```



Female patients have a no-show rate of 20.31%, compared with 19.96% for male patients. The difference is very small, suggesting that gender has little influence on appointment attendance.

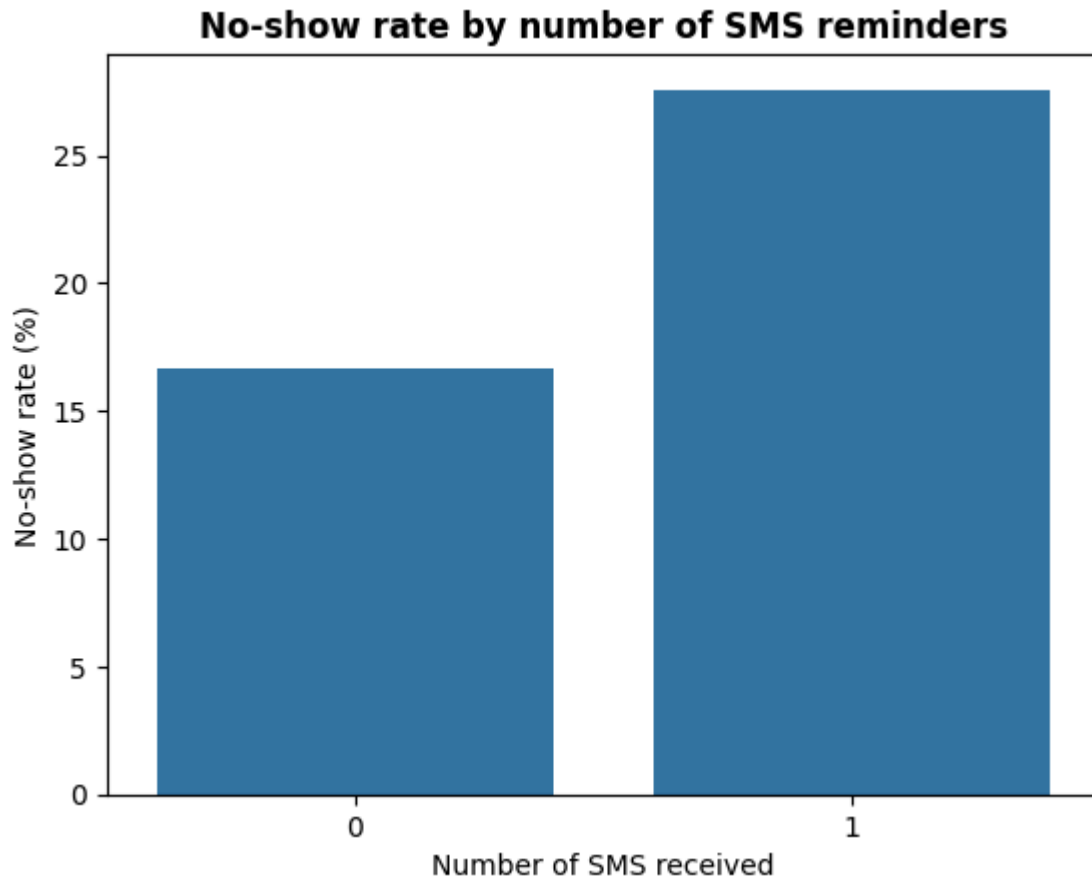


*“Do SMS reminders reduce no-shows?”*

```
sms_influence =  
df.groupby('SMS_received')['No_show_values'].mean()*100  
sms_influence
```

| SMS_received | No_show_values |
|--------------|----------------|
| 0            | 16.697984      |
| 1            | 27.574545      |

```
#Bar chart  
sns.barplot(x=sms_influence.index, y=sms_influence.values)  
sms_influence.index = sms_influence.index.map({0: "No SMS",1:  
"SMS received"})  
plt.xlabel('Number of SMS received')  
plt.ylabel('No-show rate (%)')  
plt.title('No-show rate by number of SMS reminders',  
fontweight='bold')  
plt.show()
```



The analysis shows that patients who received an SMS reminder have a higher no-show rate than those who did not receive one. However, this result should be interpreted with caution, as the dataset does not provide information about how SMS reminders were assigned.

Several factors could explain this finding. For example, patients may have been given the option to receive SMS reminders when scheduling their appointments, resulting in differences between the two groups. It is also possible that healthcare providers prioritized reminders for patients considered more likely to miss their appointments, such as those with complex medical conditions or a history of missed visits. Since this information is not available in the dataset, no causal relationship can be established between SMS reminders and appointment attendance.

*“Which neighbourhoods have the highest no-show rates?”*

```
neighbourhood_rate =  
df.groupby('Neighbourhood')['No_show_values'].mean()*100  
neighbourhood_rate
```

| Neighbourhood          | No_show_values |
|------------------------|----------------|
| AEROPORTO              | 12.500000      |
| ANDORINHAS             | 23.032714      |
| ANTÔNIO HONÓRIO        | 18.450185      |
| ARIOVALDO<br>FAVALESSA | 21.985816      |
| BARRO VERMELHO         | 21.513002      |
| ...                    | ...            |
| SÃO JOSÉ               | 21.648963      |
| SÃO PEDRO              | 21.037582      |
| TABUAZEIRO             | 18.268924      |
| UNIVERSITÁRIO          | 21.052632      |
| VILA RUBIM             | 16.568743      |

```
#Sort neighbourhoods by descending order
neighbourhood_sorted =
neighbourhood_rate.sort_values(ascending=False)
neighbourhood_sorted
```

| Neighbourhood                  | No_show_values |
|--------------------------------|----------------|
| ILHAS OCEÂNICAS DE<br>TRINDADE | 100.000000     |
| SANTOS DUMONT                  | 28.918495      |
| SANTA CECÍLIA                  | 27.455357      |
| SANTA CLARA                    | 26.482213      |

|                   |           |
|-------------------|-----------|
| ITARARÉ           | 26.266363 |
| ...               | ...       |
| SOLON BORGES      | 14.712154 |
| MÁRIO CYPRESTE    | 14.555256 |
| AEROPORTO         | 12.500000 |
| ILHA DO BOI       | 8.571429  |
| PARQUE INDUSTRIAL | 0.000000  |

We checked the number of appointments in each neighbourhood to investigate the 100% no-show rate in ILHAS OCEÂNICAS DE TRINDADE. This helped us determine whether the result was due to a limited number of observations rather than a genuine pattern.

```
df['Neighbourhood'].value_counts()
```

| Neighbourhood               | count |
|-----------------------------|-------|
| JARDIM CAMBURI              | 7717  |
| MARIA ORTIZ                 | 5805  |
| RESISTÊNCIA                 | 4430  |
| JARDIM DA PENHA             | 3877  |
| ITARARÉ                     | 3514  |
| ...                         | ...   |
| ILHA DO BOI                 | 35    |
| ILHA DO FRADE               | 10    |
| AEROPORTO                   | 8     |
| ILHAS OCEÂNICAS DE TRINDADE | 2     |
| PARQUE INDUSTRIAL           | 1     |

The neighbourhood ILHAS OCEÂNICAS DE TRINDADE shows a 100% no-show rate. However, this neighbourhood contains only two appointments, making the result unreliable. We therefore excluded it from the analysis.

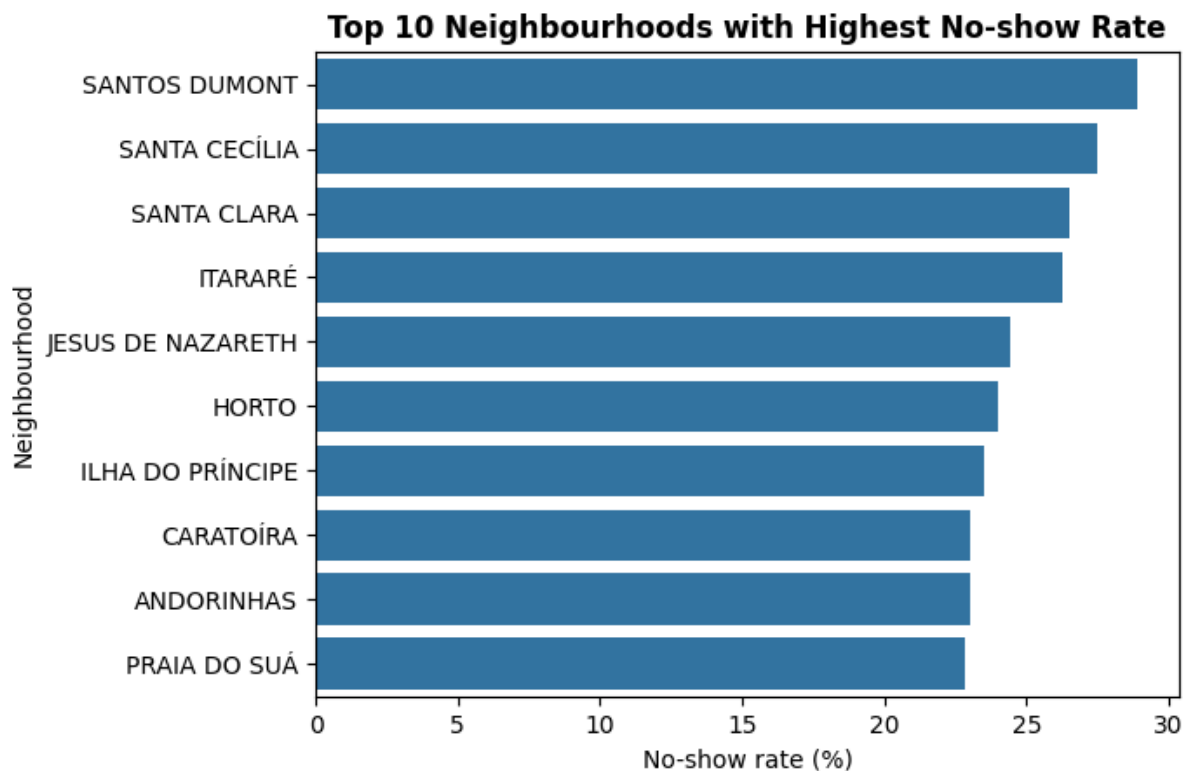
```
#Drop ILHAS OCEÂNICAS DE TRINDADE
neighbourhood_rate =
neighbourhood_rate[df['Neighbourhood'].value_counts() > 30]
```

```
#Top 10 neighbourhoods with the highest no show rate
top_10 =
neighbourhood_rate.sort_values(ascending=False).head(10)
top_10
```

| Neighbourhood     | No_show_values |
|-------------------|----------------|
| SANTOS DUMONT     | 28.918495      |
| SANTA CECÍLIA     | 27.455357      |
| SANTA CLARA       | 26.482213      |
| ITARARÉ           | 26.266363      |
| JESUS DE NAZARETH | 24.395373      |
| HORTO             | 24.000000      |
| ILHA DO PRÍNCIPE  | 23.477493      |
| CARATOÍRA         | 23.040936      |
| ANDORINHAS        | 23.032714      |
| PRAIA DO SUÁ      | 22.826087      |

```
#Bar plot
sns.barplot(x=top_10.values, y=top_10.index)
plt.xlabel('No-show rate (%)')
plt.ylabel('Neighbourhood')
plt.title("Top 10 Neighbourhoods with Highest No-show Rate",
fontweight='bold')
```

```
plt.show()
```



The analysis reveals noticeable differences in no-show rates across neighbourhoods. Santos Dumont, Santa Cecília, Santa Clara and Itararé exhibit the highest absenteeism rates, all exceeding 26%.

These results suggest that geographical location may play a role in appointment attendance. However, the dataset does not provide information on socioeconomic factors, transportation access, or proximity to healthcare facilities. Therefore, while neighbourhood appears to be associated with no-show behavior, the underlying reasons for these differences cannot be determined from the available data.

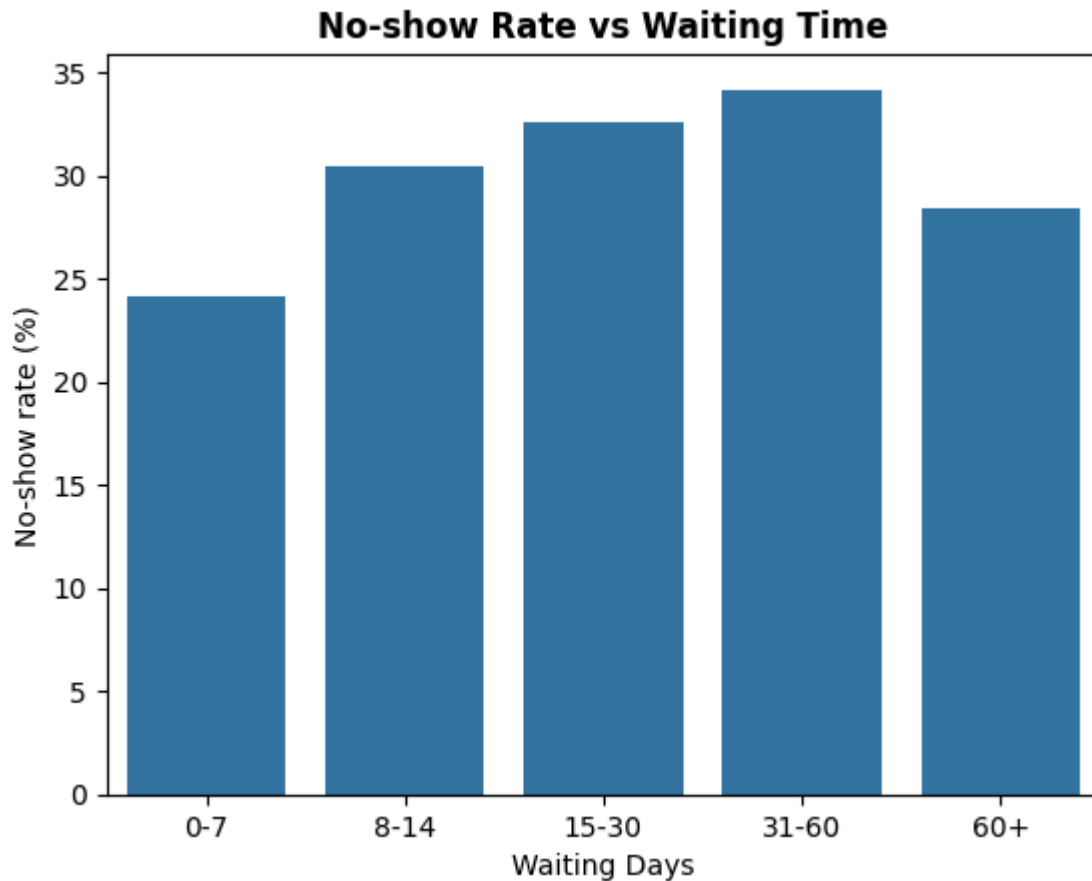
*“Does waiting time between scheduling and appointment affect attendance?”*

```
#Group waiting days by buckets
df['waiting_group'] = pd.cut(df['Waiting Days'], bins=[0, 7,
14, 30, 60, 180], labels=['0-7', '8-14', '15-30',
'31-60', '60+'])
```

```
waitingtime_influence =  
df.groupby('waiting_group')['No_show_values'].mean()*100  
waitingtime_influence
```

|               | No_show_values |
|---------------|----------------|
| waiting_group |                |
| 0-7           | 24.147895      |
| 8-14          | 30.469854      |
| 15-30         | 32.588797      |
| 31-60         | 34.154292      |
| 60+           | 28.448687      |

```
#Barplot  
sns.barplot(x=waitingtime_influence.index,  
y=waitingtime_influence.values)  
plt.xlabel("Waiting Days")  
plt.ylabel('No-show rate (%)')  
plt.title('No-show Rate vs Waiting Time', fontweight= 'bold')  
plt.show()
```



The analysis shows that no-show rates generally increase as the waiting time between booking and appointment increases. Patients with longer delays tend to miss their appointments more frequently than those with shorter waiting times.

One possible explanation is that patients may no longer perceive the appointment as necessary, may have rescheduled their care elsewhere, or may simply forget about their appointment as the waiting time increases. However, these remain hypotheses, as the dataset does not provide information on patient intentions or external scheduling behavior.

However, this trend is not strictly linear. A slight decrease is observed for the longest waiting period (60+ days), which may suggest that patients who wait very long for an appointment are either more committed to attending or represent a specific subgroup of patients with different healthcare needs.

Overall, waiting time appears to be an important factor associated with appointment attendance, although the relationship is not perfectly monotonic.

*“Do days of the week affect attendance?”*



```

day_influence = df.groupby('Appointment
day')['No_show_values'].mean()*100
day_influence

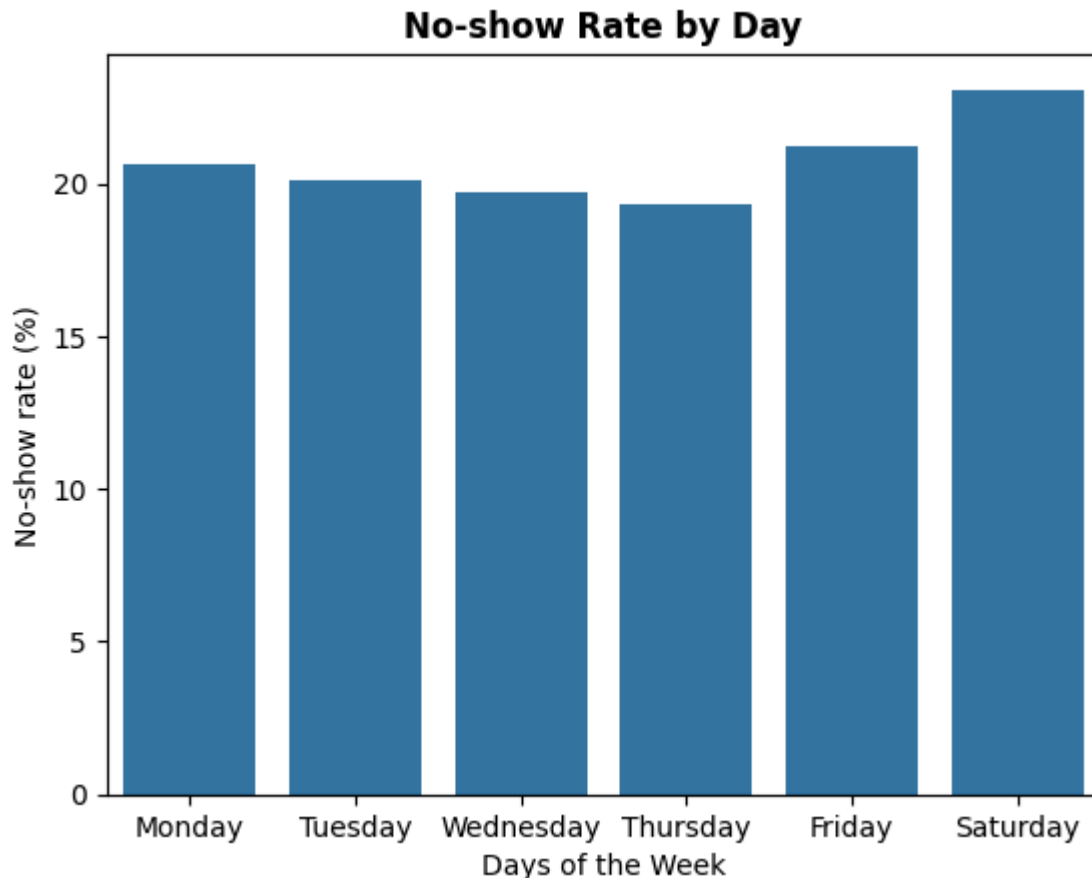
```

| Appointment<br>day | No_show_values |
|--------------------|----------------|
| Friday             | 21.226142      |
| Monday             | 20.644565      |
| Saturday           | 23.076923      |
| Thursday           | 19.349414      |
| Tuesday            | 20.087370      |
| Wednesday          | 19.686074      |

```

#Barplot
day_order = 'Monday', 'Tuesday', 'Wednesday', 'Thursday',
'Friday', 'Saturday'
sns.barplot(x=day_influence.index, y=day_influence.values,
order= day_order )
plt.xlabel("Days of the Week")
plt.ylabel('No-show rate (%)')
plt.title('No-show Rate by Day', fontweight= 'bold')
plt.show()

```



The analysis shows relatively small variations in no-show rates across weekdays. Mid-week days such as Wednesday and Thursday tend to have slightly lower absenteeism rates, while Friday and especially Saturday show higher no-show rates.

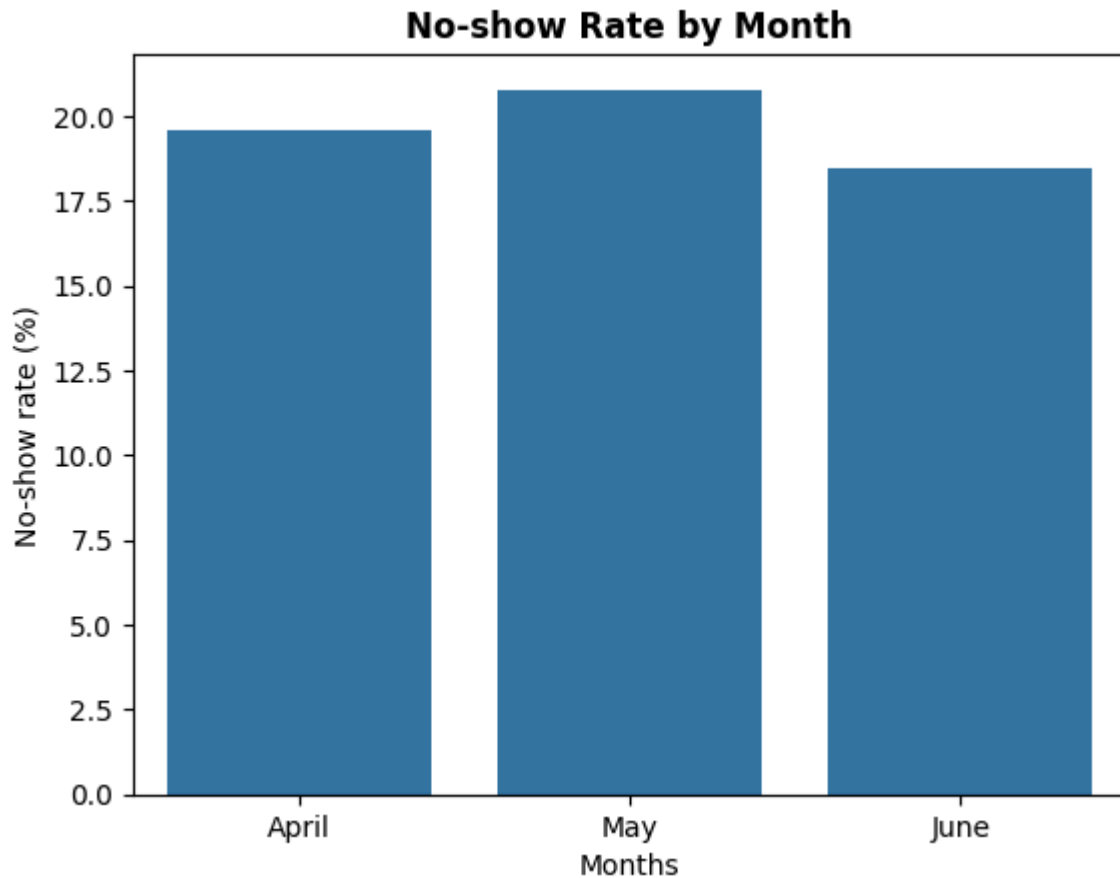
However, the differences between weekdays are not very large, suggesting that the day of the week may have only a limited influence on appointment attendance compared to other factors such as waiting time or patient characteristics.

*“Do months affect attendance?”*

```
#No show by month
month_influence = df.groupby('Appointment
month')['No_show_values'].mean()*100
month_influence
```

|                   | No_show_values |
|-------------------|----------------|
| Appointment month |                |
| April             | 19.567233      |
| June              | 18.457467      |
| May               | 20.781582      |

```
#Barplot
month_order = 'April', 'May', 'June'
sns.barplot(x=month_influence.index, y=month_influence.values,
order= month_order )
plt.xlabel("Months")
plt.ylabel('No-show rate (%)')
plt.title('No-show Rate by Month', fontweight= 'bold')
plt.show()
```



**The absenteeism rate remains relatively stable across April, May, and June, with only minor variations (20% in May versus 18% in June).**

**These differences are not large enough to suggest a strong seasonal effect on appointment attendance within the observed period.**

## **KEY INSIGHTS**

- This analysis shows that approximately 80% of patients attend their appointments, while around 20% result in no-shows.
- Age appears to be an important factor: younger patients (0–30 years old) have higher no-show rates, which gradually decrease with age.
- No clear relationship was found between no-show rates and gender or the receipt of SMS reminders, suggesting that these factors alone may not significantly influence attendance.
- Geographical location also shows variation in no-show rates across neighbourhoods, which may indicate an association between location and attendance behavior. However, the underlying causes cannot be determined from this dataset.
- Waiting time between booking and appointment shows a positive association with no-show rates: longer delays tend to correspond to higher absenteeism.
- The day of the week shows only slight variation, with lower no-show rates observed in the middle of the week and slightly higher rates toward the end of the week (Friday and Saturday).
- Overall, no-shows appear to be influenced more by patient-related factors (such as age) and waiting time than by calendar-based variables such as month or weekday.

## **RECOMMENDATIONS**

- Implement appointment confirmation reminders (SMS or phone call) the day before the appointment to reduce forgetfulness and improve attendance rates.
- Introduce a follow-up policy for missed appointments (e.g., tracking repeated no-shows or implementing rescheduling support) to better manage patient attendance behavior.
- Offer a waitlist or notification system to inform patients when earlier appointment slots become available due to cancellations, helping reduce long waiting times.
- Develop telemedicine options for patients facing mobility, distance, or scheduling constraints, which could help reduce no-show rates for specific patient groups.

# **CONCLUSION**

This analysis explored factors associated with medical appointment no-shows using demographic, scheduling, and contextual variables.

The results show that no-show behavior is mainly influenced by patient age and waiting time. Younger patients tend to miss more appointments, while longer delays between booking and the appointment date are associated with higher absenteeism rates. In contrast, variables such as gender, month, and SMS reminders show little or no clear impact on attendance.

Geographical differences between neighbourhoods suggest some variation in no-show behavior, although the underlying causes cannot be determined from the available data.

Overall, the analysis indicates that no-shows are more strongly related to patient characteristics and scheduling delays than to calendar-based factors.

However, this study is limited to observational data, meaning that no causal relationships can be established.